OULAD Individual Submission

Part 1: Notebook Link

Notebook filename: OULAD_Individual_2550723.ipynb

Notebook link: https://github.com/GIft-Vundla/oulad-week4-early-outcome-prediction/blob/main/OULAD_Individual_2550723.ipynb

Part 2: Answers To Questions

2.1 What procedure did you take to avoid leakage?

Leakage was controlled by enforcing the observation window before any modelling. Only evidence available by day 27 was retained, post-day-27 VLE and assessment information was excluded, negative pre-start clicks were not modeled, exact duplicate week-4 VLE rows were removed, and date_unregistration and assessment score were excluded. After the feature table was built, the data was split into train, validation, and test sets. All imputations, encoding, and scaling in the logistic pipeline were fit on training data only. CatBoost class weighting was learned on training data only, thresholds were selected on validation data only, and the test split was used once for final reporting. This ensured that neither future outcomes nor information from validation or test data influenced model fitting. See Figure 1 and Table 4.

2.2 How did you handle missing data?

Missing data was handled according to meaning rather than with a single global rule. Structural missingness occurs for students with no valid week-4 activity, producing 4,707 missing values in first_activity_day_w4 and 4,707 in days_since_last_activity_w4; these were left missing because they indicate inactivity. Count-based activity and assessment aggregates were zero-filled so inactive students stayed in the feature table. True source missingness remained in fields such as imd_band (1,111) and date_registration (45). The logistic pipeline then applied train-only imputation: median for numeric variables and most-frequent imputation for categorical variables. CatBoost handled missing categorical values through an explicit Missing category. No information from validation or test data was used during imputation. See Table 3 and Table 4.

2.3 How did you handle outliers?

Outliers were assessed on numeric features using the 1.5 IQR rule, and the counts were recorded in the feature catalog. Extreme click totals, active-day counts, and assessment-weight features were retained because, in this context, they are plausible behavioural observations rather than obvious errors. Removing or winsorising them would risk discarding informative engagement patterns. Robustness was handled through modelling choices instead: StandardScaler was used for logistic regression, and the advanced model was tree-based CatBoost, which is less sensitive to monotonic scaling issues. Values were excluded only when they violated the study design, such as activity outside day 0-27, because those rows are not valid observations for the prediction window. See Table 4.

2.4 How did you handle duplicates?

Duplicates were checked at the appropriate key for each source table. studentInfo, studentRegistration, vle, assessments, and studentAssessment had zero duplicates at their natural keys. For studentVle, the exact event-row key (code_module, code_presentation, id_student, id_site, date, sum_click) was checked within the modeled week-4 window, and 140,592 exact duplicates were found. Those rows were removed before aggregation so week-4 click totals, active-day counts, and activity-type features were not double counted. This was important because duplicated event rows would otherwise inflate both engagement volume and derived ratios. See Table 2 and Table 4.

2.5 Explain the performance difference between your two pipelines

The logistic baseline performed well because logistic regression is a credible and interpretable baseline for dropout prediction on institutional tabular data [1], and the week-4 feature table already contains strong early-warning signal. CatBoost improved the probability model by capturing nonlinear effects and interactions across mixed categorical and numerical features, which is consistent with education-specific CatBoost evidence and the wider CatBoost literature [2], [3]. On the test set, raw CatBoost improved log loss from 0.5092 to 0.4989 and ROC-AUC from 0.8185 to 0.8264. Raw CatBoost is therefore the best probability model. Calibration did not improve log loss in this dataset, but at threshold 0.3 calibrated CatBoost reached recall 0.9145 and F1 0.7614, making it the strongest intervention-oriented configuration. Calibration was treated separately because post-hoc probability calibration is supported by prior work on classifier miscalibration [4], while the lower threshold was chosen on the validation split because early-warning decisions should be guided by recall and specificity trade-offs rather than a fixed 0.5 rule [5]. See Figure 7, Figure 8, and Table 5.

2.6 Briefly describe the ethical aspects and issues in this project

A false positive could label a student as at risk when they would have passed, which may cause unnecessary intervention or stigma. A false negative could miss a student who needs timely support. One misuse scenario would be using the model for punitive ranking or automatic exclusion instead of supportive outreach. A fair concern is that demographic variables such as age band, disability status, region, or deprivation band can reflect structural inequality, so model outputs must not be treated as neutral truth [6]. The model should support human judgment, privacy protection, and proportionate intervention. AI assistance disclosure: AI tools were used for grammar, spelling and formatting.

Part 3: Figures

Methodology block diagram

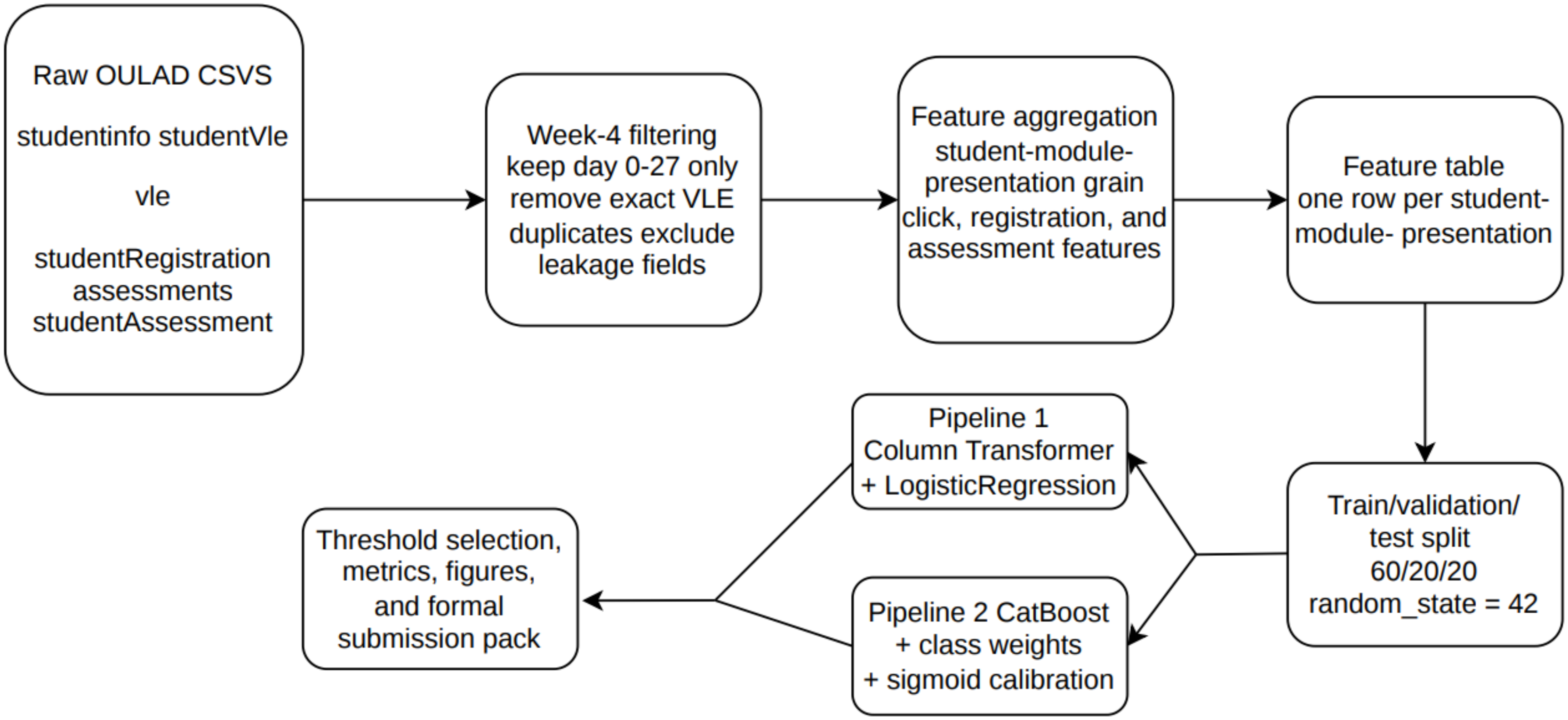

Figure 1: End-to-end workflow for the formal brief: raw OULAD tables are filtered to the week-4 window, exact duplicate VLE event rows are removed, features are aggregated to student-module-presentation level, the data is split into train/validation/test sets, and the two pipelines are evaluated before threshold selection and formal artifact export.

total_clicks_w4 by binary target

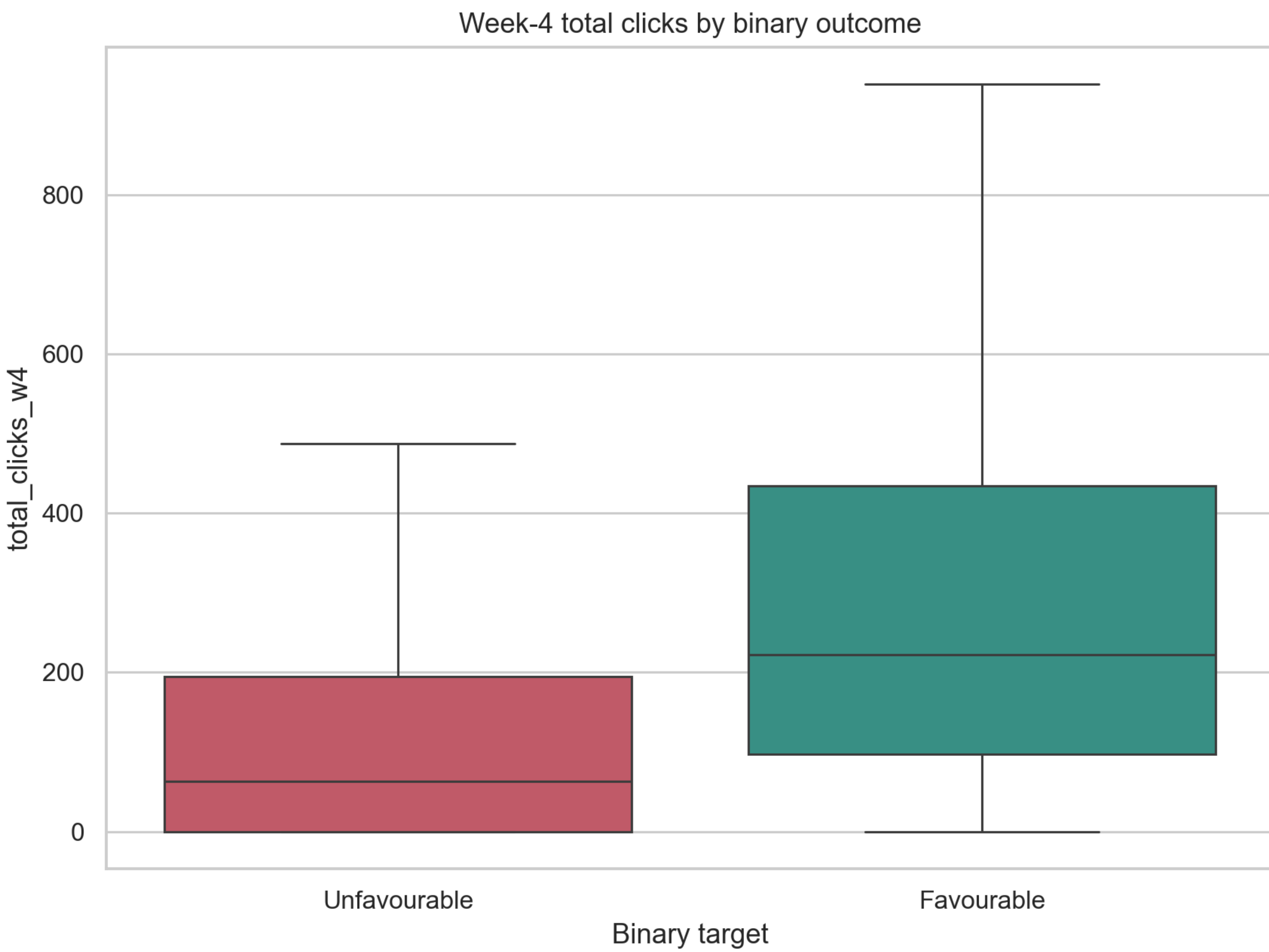

Figure 2: Students in the favourable class tend to accumulate higher total click volume by week 4 than students in the unfavourable class, indicating a positive relationship between early VLE engagement and outcome.

active_days_w4 by binary target

Figure 3: Favourable outcomes are associated with activity spread across more distinct days in the week-4 window, suggesting that consistency of engagement matters in addition to raw click volume.

clicks_per_active_day_w4 by binary target

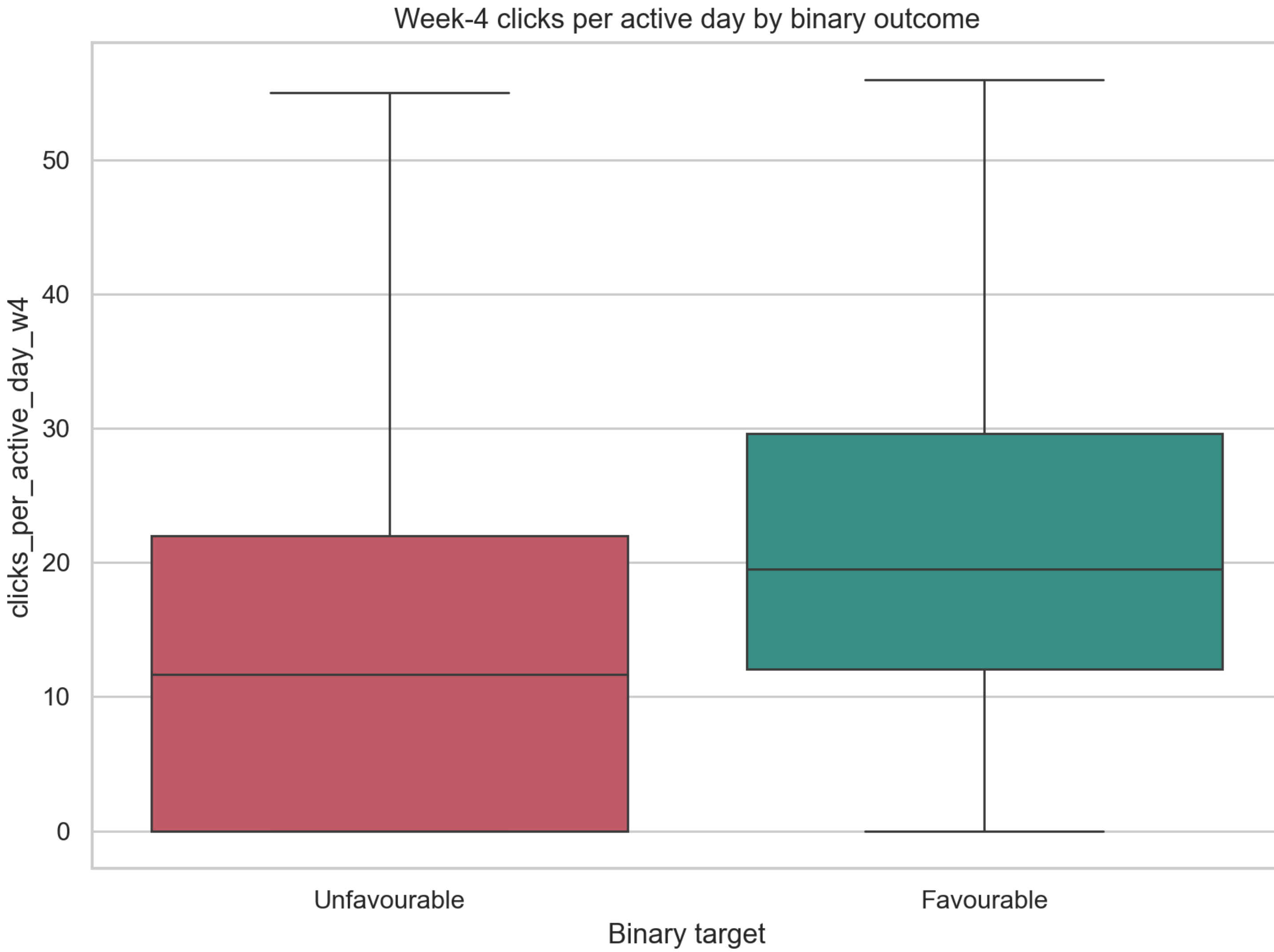

Figure 4: Favourable outcomes are associated with stronger engagement intensity on days when students are active, which supports using an intensity ratio rather than only total clicks.

weighted_submitted_by_w4 by binary target

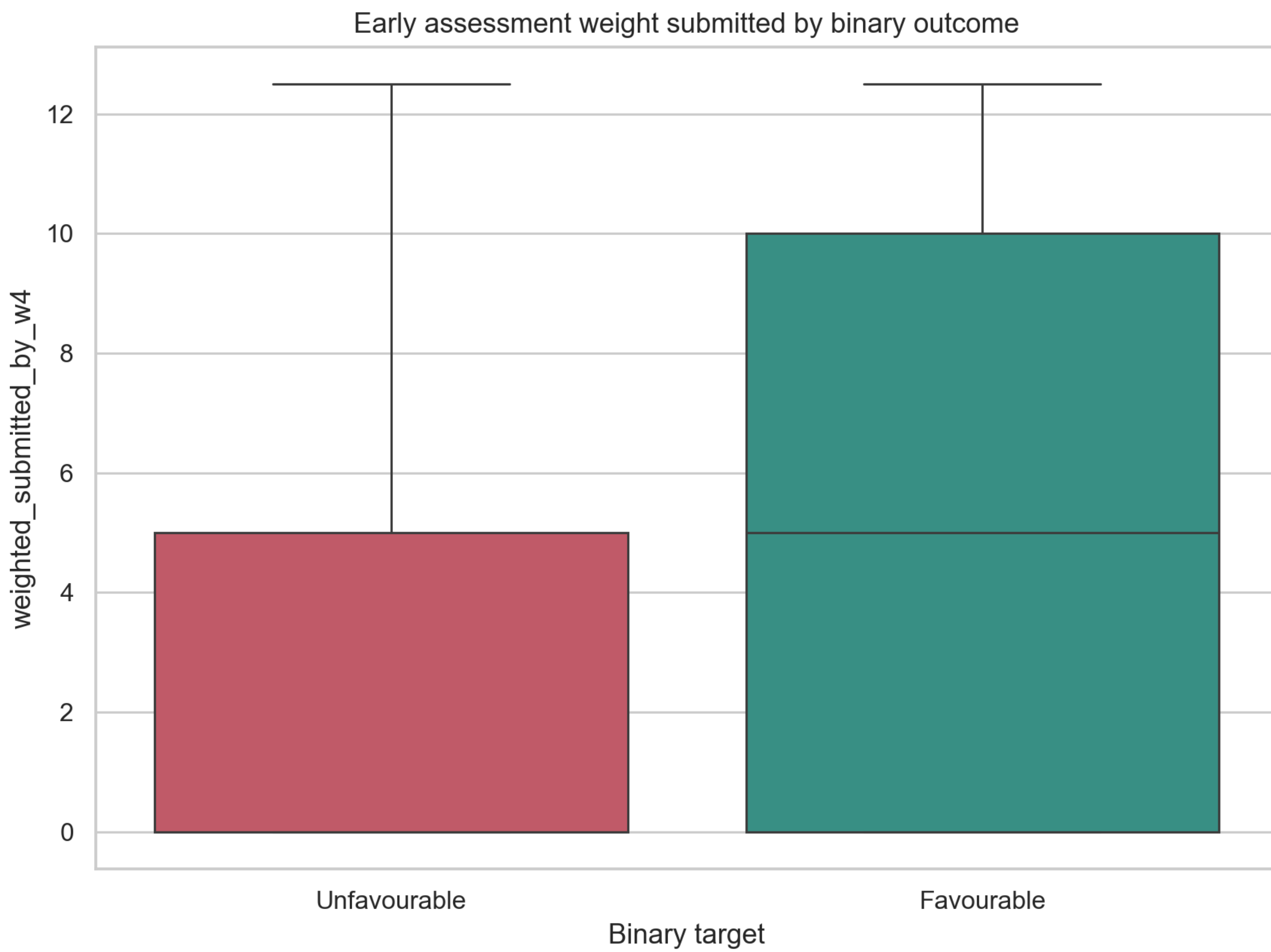

Figure 5: Students in the favourable class generally submit more assessment weight by week 4, showing that early coursework behaviour carries positive signal for final outcome.

Confusion matrices for the four reported configurations

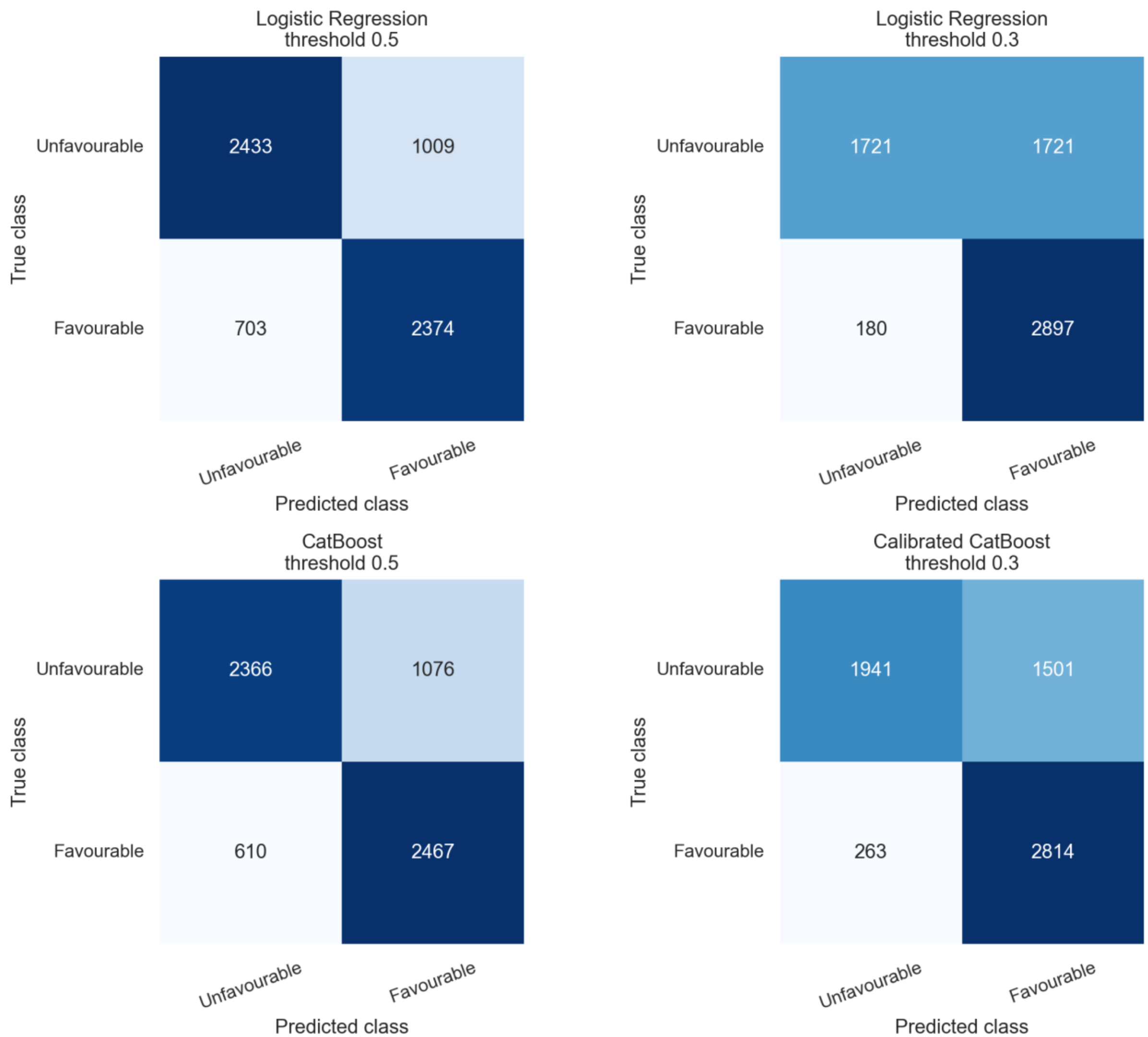

Figure 6. Held-out test-split confusion matrices for P1 logistic regression at threshold 0.5, P1-T logistic regression at threshold 0.3, P2 raw CatBoost at threshold 0.5, and P2-T calibrated CatBoost at threshold 0.3. The tuned 0.3-threshold configurations reduce false negatives substantially, but this comes at the cost of more false positives for the favourable-versus-unfavourable week-4 prediction task.

ROC comparison on the test split

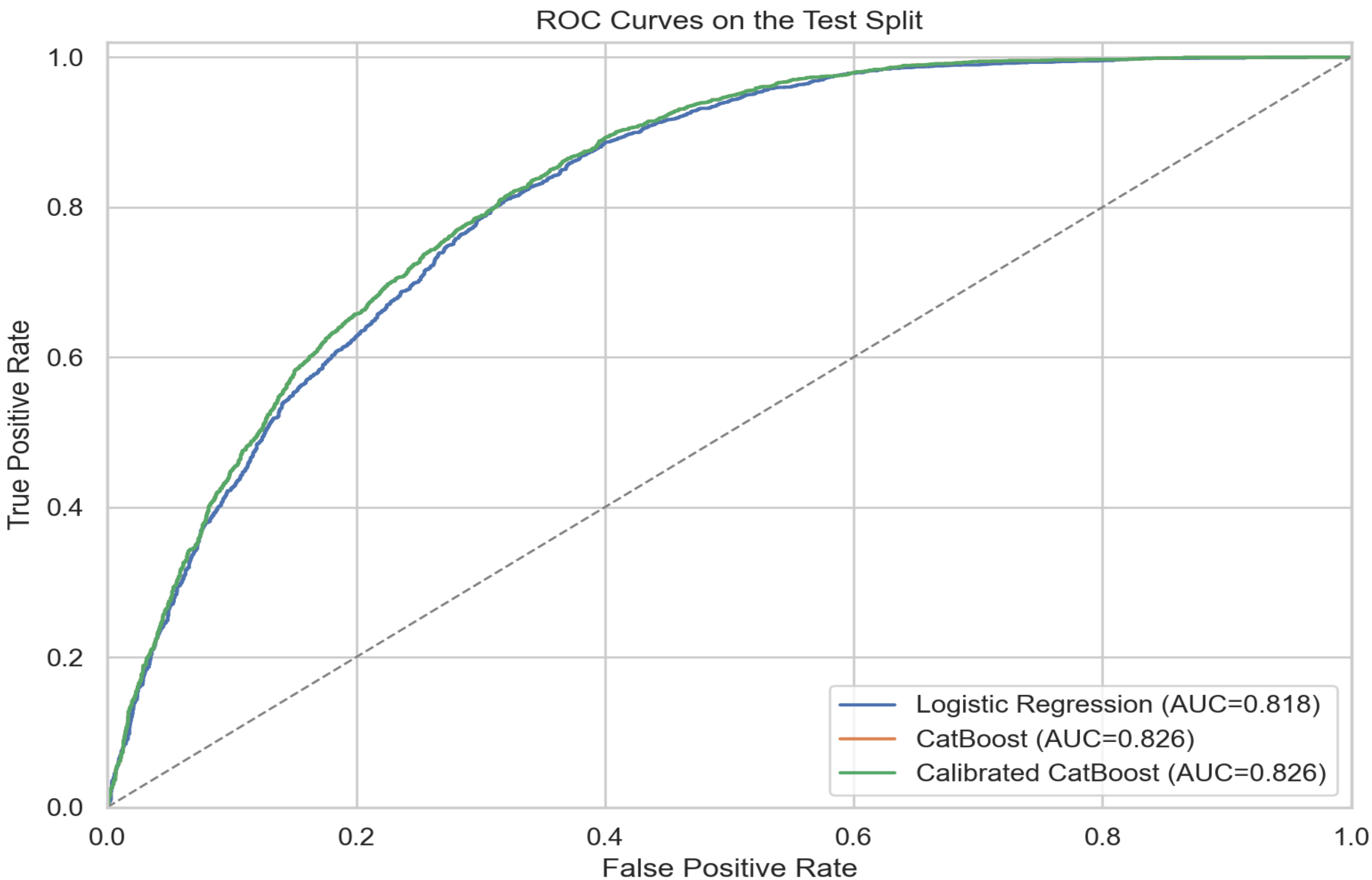

Figure 7: ROC curves on the held-out test split for the week-4 favourable-versus-unfavourable prediction task. Raw CatBoost maintains the strongest probability-ranking performance, with the best ROC-AUC among the reported configurations.

Figure 8. Threshold sweep for logistic regression and calibrated CatBoost

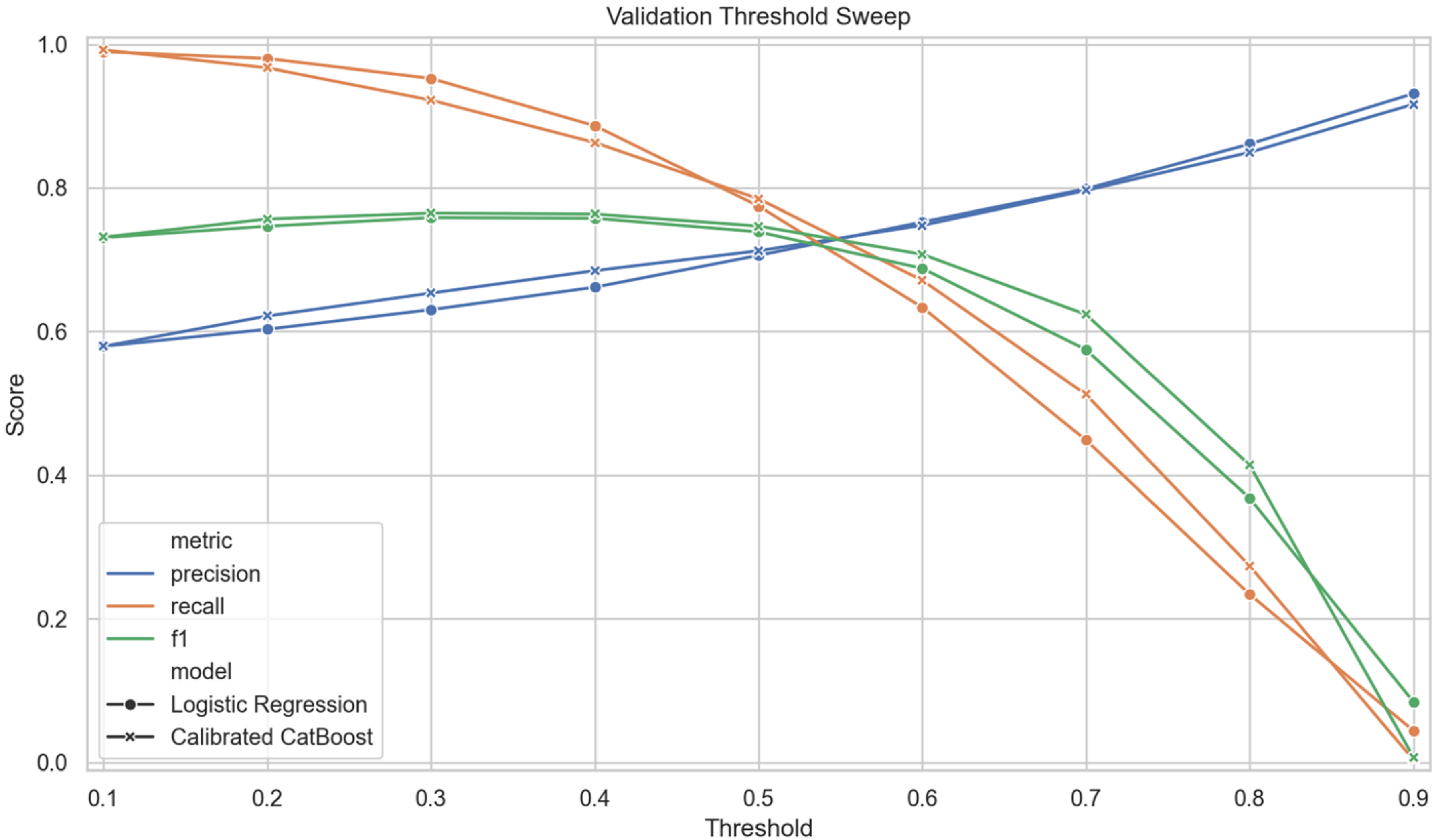

Figure 8. Validation-split threshold sweeps for logistic regression and calibrated CatBoost across thresholds from 0.1 to 0.9. Both models achieve their best validation F1 at threshold 0.3, which justifies using a lower intervention threshold than the default 0.5 before locking the final test-set comparison.

Grouped metric comparison

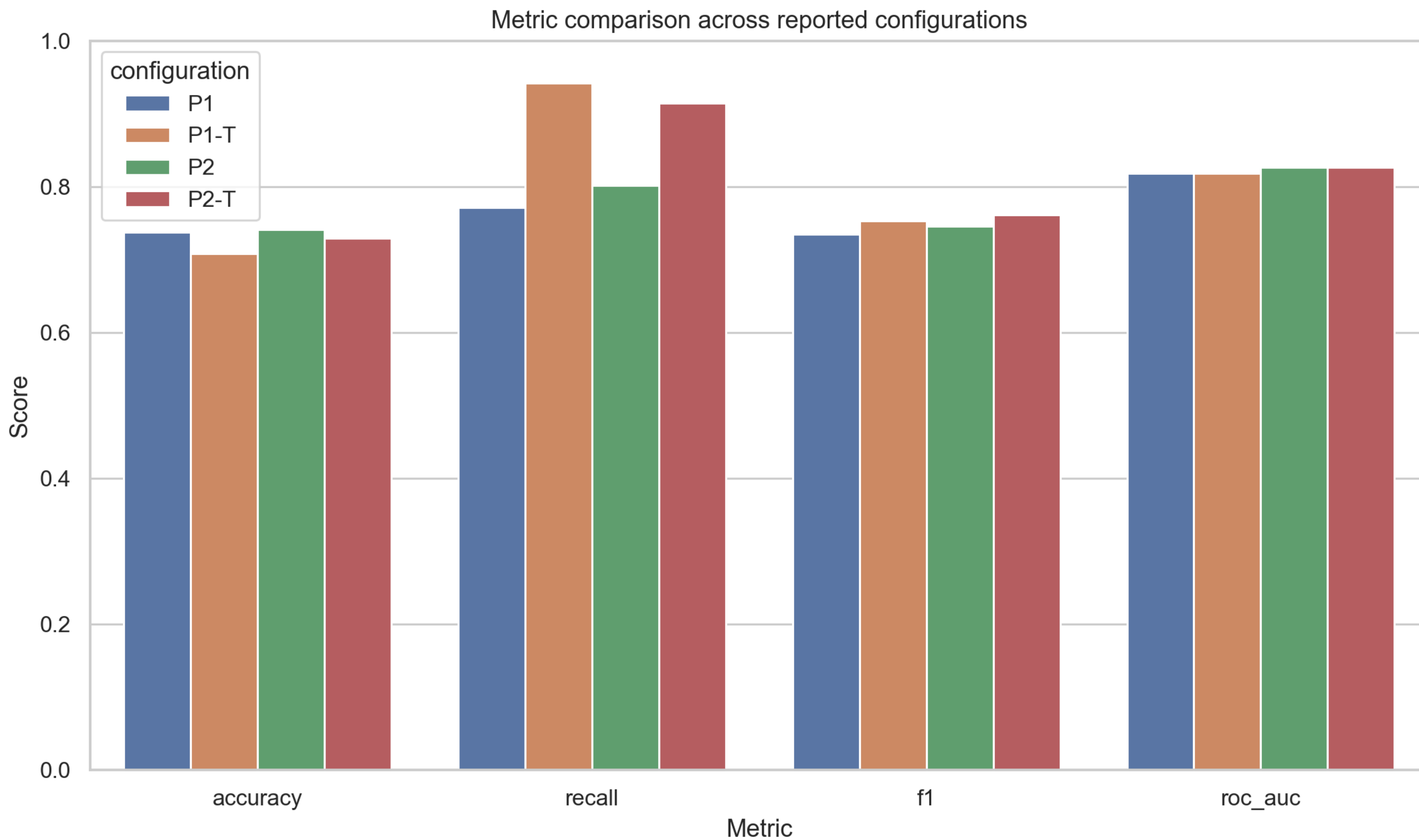

Figure 9. Raw CatBoost is the strongest probability model on log loss and ROC-AUC, while calibrated CatBoost at threshold 0.3 provides the best intervention-oriented recall and F1 balance.

Part 4: Tables

Table 1: Binary label mapping and class balance

Binary target	Original OULAD values	Count	Percentage
Favourable	Pass, Distinction	15385	47.2%
Unfavourable	Fail, Withdrawn	17208	52.8%

Table 1: The target mapping treats Pass and Distinction as favourable and Fail and Withdrawn as unfavourable, which satisfies the brief's favourable-versus-unfavourable framing derived from final_result.

Table 2. Row counts for loaded source tables and filtered subsets

Table or derived subset	Rows
studentInfo.csv	32593
studentRegistration.csv	32593
vle.csv	6364
studentVle.csv	10655280
studentVle rows within day 0-27	2112323
Exact duplicate studentVle rows removed before aggregation	140592
assessments.csv	206
Assessments due by day 27	17
studentAssessment.csv	173912
Assessment submissions by day 27	20739

Table 2: This table documents the size of each source table, the week-4 filtered event subset, and the exact duplicate VLE rows removed before aggregation.

Table 3: Missing-value summary for modeled columns with non-zero missingness

Column	Missing count
days_since_last_activity_w4	4707
first_activity_day_w4	4707
imd_band	1111
date_registration	45

Table 3. Missingness is concentrated in structural inactivity fields and a small number of true demographic or registration fields; zero-filled aggregates and train-only imputation were used accordingly.

Table 4 is intentionally wide because it follows the brief's required metadata columns. It is rendered in landscape orientation here.

Table 4: Feature catalog for the modeled week-4 feature table

Feature	Source CSV(s)	Original column(s) used	Data type	Definition	How computed	Week availability (2/4/6/8)	Missing count	Outliers count	Duplicate count	Leakage risk	Notes
code_module	studentInfo.csv	code_module	cat	Module identifier.	as-is	2/4/6/8	0		0		Static course identifier.
code_presentation	studentInfo.csv	code_presentation	cat	Presentation identifier.	as-is	2/4/6/8	0		0		Static presentation identifier.
gender	studentInfo.csv	gender	cat	Student gender category.	as-is	2/4/6/8	0		0	Low	Fairness-sensitive demographic field.
region	studentInfo.csv	region	cat	Student region category.	as-is	2/4/6/8	0		0	Low	Fairness-sensitive demographic field.
highest_education	studentInfo.csv	highest_education	cat	Highest education band.	as-is	2/4/6/8	0		0	Low	Static background feature.
imd_band	studentInfo.csv	imd_band	cat	Socioeconomic deprivation band.	as-is	2/4/6/8	1111		0	Low	Fairness-sensitive socioeconomic proxy.
age_band	studentInfo.csv	age_band	cat	Age category.	as-is	2/4/6/8	0		0	Low	Static demographic band.

num_of_prev_attempts	studentInfo.csv	num_of_prev_attempts	num	Number of prior attempts on the module.	as-is	2/4/6/8	0	4172.0	0		Static history feature.
studied_credits	studentInfo.csv	studied_credits	num	Number of studied credits.	as-is	2/4/6/8	0	350.0	0		Static workload feature.
disability	studentInfo.csv	disability	cat	Disability disclosure flag.	as-is	2/4/6/8	0		0	Low	Fairness-sensitive field.
date_registration	studentRegistration.csv	date_registration	num	Registration day relative to presentation start.	as-is	2/4/6/8	45	339.0	0		Known before or at course start.
registered_before_start_days	studentRegistration.csv	date_registration	num	Days registered before day 0.	max(0, -date_registration)	2/4/6/8	0	332.0	0		Derived from registration timing only.
late_registration_flag	studentRegistration.csv	date_registration	bool	Whether registration happened after day 0.	1 if date_registration > 0 else 0	2/4/6/8	0		0		Binary timing flag.
total_clicks_w4	studentVle.csv	date, sum_click	num	Total VLE clicks observed by week 4.	sum(sum_click) where 0 <= date <= 27	4/6/8	0	1919.0	140592	Low	Week-4 aggregate after duplicate removal.
first_activity_day_w4	studentVle.csv	date	num	First valid activity day in the week-4 window.	min(date) where 0 <=	4/6/8	4707	2253.0	140592	Low	Missing for inactive students.

					date <= 27						
active_days_w4	studentVle.csv	date	num	Number of distinct active click days by week 4.	nunique(date) where 0 <= date <= 27	4/6/8	0	0.0	140592	Low	Zero-filled for inactive students.
unique_sites_w4	studentVle.csv	id_site	num	Number of distinct VLE sites visited by week 4.	nunique(id_site) where 0 <= date <= 27	4/6/8	0	689.0	140592	Low	Site breadth feature.
recommended_clicks_w4	studentVle.csv, vle.csv	date, sum_click, recommended	num	Clicks on resources flagged as recommended.	sum(sum_click) on recommended resources where 0 <= date <= 27	4/6/8	0	0.0	140592	Low	Workspace `vle.csv` lacks `recommended`; values are zero placeholders.
forumng_clicks_w4	studentVle.csv, vle.csv	date, sum_click, activity_type	num	Week-4 clicks on forumng resources.	sum(sum_click) where activity_type = 'forumng' and 0 <= date <= 27	4/6/8	0	2799.0	140592	Low	Activity-type engagement feature.

homepage_clicks_w4	studentVle.csv, vle.csv	date, sum_click, activity_type	num	Week-4 clicks on homepage resources.	sum(sum_click) where activity_type = 'homepage' and 0 <= date <= 27	4/6/8	0	1839.0	140592	Low	Activity-type engagement feature.
other_activity_clicks_w4	studentVle.csv, vle.csv	date, sum_click, activity_type	num	Week-4 clicks on other activity resources.	sum(sum_click) where activity_type = 'other_activity' and 0 <= date <= 27	4/6/8	0	5252.0	140592	Low	Activity-type engagement feature.
oucontent_clicks_w4	studentVle.csv, vle.csv	date, sum_click, activity_type	num	Week-4 clicks on oucontent resources.	sum(sum_click) where activity_type = 'oucontent' and 0 <= date <= 27	4/6/8	0	3489.0	140592	Low	Activity-type engagement feature.
page_clicks_w4	studentVle.csv, vle.csv	date, sum_click, activity_type	num	Week-4 clicks on page resources.	sum(sum_click) where	4/6/8	0	4965.0	140592	Low	Activity-type

					activity_type = 'page' and 0 <= date <= 27						engagement feature.
quiz_clicks_w4	studentVle.csv, vle.csv	date, sum_click, activity_type	num	Week-4 clicks on quiz resources.	sum(sum_click) where activity_type = 'quiz' and 0 <= date <= 27	4/6/8	0	3029.0	140592	Low	Activity-type engagement feature.
resource_clicks_w4	studentVle.csv, vle.csv	date, sum_click, activity_type	num	Week-4 clicks on resource resources.	sum(sum_click) where activity_type = 'resource' and 0 <= date <= 27	4/6/8	0	1611.0	140592	Low	Activity-type engagement feature.
subpage_clicks_w4	studentVle.csv, vle.csv	date, sum_click, activity_type	num	Week-4 clicks on subpage resources.	sum(sum_click) where activity_type = 'subpage' and 0 <= date <= 27	4/6/8	0	2169.0	140592	Low	Activity-type engagement feature.

url_clicks_w4	studentVle.csv, vle.csv	date, sum_click, activity_type	num	Week-4 clicks on url resources.	sum(sum_click) where activity_type = 'url' and 0 <= date <= 27	4/6/8	0	3038.0	140592	Low	Activity-type engagement feature.
clicks_per_active_day_w4	studentVle.csv	date, sum_click	num	Average clicks per active day by week 4.	total_clicks_w4 / active_days_w4	4/6/8	0	1005.0	140592	Low	Intensity measure.
days_since_last_activity_w4	studentVle.csv	date	num	Recency of the last valid week-4 activity.	27 - max(date) where 0 <= date <= 27	4/6/8	4707	2285.0	140592	Low	Missing for inactive students.
recommended_click_ratio_w4	studentVle.csv, vle.csv	date, sum_click, recommended	num	Share of total week-4 clicks on recommended resources.	recommended_clicks_w4 / total_clicks_w4	4/6/8	0	0.0	140592	Low	Workspace `vle.csv` lacks `recommended`; values are zero placeholders.
early_assessment_due_count	assessments.csv	id_assessment, date	num	Number of assessments due by day 27 for the	count distinct id_assessment	4/6/8	0	6771.0	0	Low	Module-presentation level feature.

				module presentation.	where 0 <= date <= 27						
weighted_due_by_w4	assessments.csv	date, weight	num	Total assessment weight due by day 27.	sum(weight) where 0 <= date <= 27	4/6/8	0	0.0	0	Low	Module-presentation level feature.
early_assessment_submitted_count	assessments.csv, studentAssessment.csv	id_assessment, id_student, date_submitted	num	Number of early assessments submitted by day 27.	count distinct id_assessment after joining early assessments and submissions	4/6/8	0	0.0	0	Low	Submission behavior feature.
on_time_submission_count	assessments.csv, studentAssessment.csv	date, date_submitted	num	Count of early assessments submitted on or before the due date.	sum(1 where date_submitted - due_date <= 0)	4/6/8	0	0.0	0	Low	Submission punctuality feature.
late_submission_count	assessments.csv, studentAssessment.csv	date, date_submitted	num	Count of early assessments submitted late but by day 27.	sum(1 where date_submitted - due_date > 0)	4/6/8	0	5223.0	0	Low	Submission punctuality feature.

mean_submission_delay_days	assessments.csv, studentAssessment.csv	date, date_submitted	num	Average submission delay for early assessments.	mean(date_submitted - due_date)	4/6/8	0	8389.0	0	Low	Zero-filled when no early submissions exist.
weighted_submitted_by_w4	assessments.csv, studentAssessment.csv	weight, date_submitted	num	Total assessment weight submitted by day 27.	sum(weight) for early assessments submitted by day 27	4/6/8	0	0.0	0	Low	Strong early coursework signal.
submitted_any_early_assessment	assessments.csv, studentAssessment.csv	id_assessment, date_submitted	bool	Whether any early assessment was submitted by day 27.	1 if early_assessment_submitted_count > 0 else 0	4/6/8	0		0	Low	Binary early-engagement flag.

Table 4: The feature catalog records week availability, missingness, outlier counts, duplicate counts, leakage risk, and notes for every modeled feature. For VLE-derived features, duplicate counts refer to exact week-4 event-row duplicates removed before aggregation. The local vle.csv does not contain the optional recommended field, so recommended-resource features remain zero-valued placeholders.

Table 5: Pipeline comparison table

Pipeline ID	Feature set	Encoding	Scaling	Model	Evaluation method	Hyperparameters	Seed	Chosen threshold	Accuracy	Precision	Recall	Specificity	FPR	F1	Log loss	AUC
P1	All week-4 features	Most-frequent imputation + one-hot categorical encoding	Standard Scaler on numeric columns only	LogisticRegression(max_iter=1000)	Split 60/20/20 with threshold selection on validation; 5-fold CV summary on training	max_iter=1000, random_state=42	42	0.5	0.7374	0.7017	0.7715	0.7069	0.2931	0.735	0.5092	0.8185
P1-T	All week-4 features	Most-frequent imputation + one-hot categorical encoding	Standard Scaler on numeric columns only	LogisticRegression(max_iter=1000)	Split 60/20/20 with threshold selection on validation; 5-fold CV	max_iter=1000, random_state=42	42	0.3	0.7084	0.6273	0.9415	0.5	0.5	0.753	0.5092	0.8185

					summary on training											
P2	All week-4 features	Native categorical handling with explicit Missing category		CatBoostClassifier	Split 60/20/20 with threshold selection on validation; 5-fold CV summary on training	loss_function=Logloss, eval_metric=AUC, depth=6, learning_rate=0.05, iterations=500, auto_class_weights=Balanced, random_state=42	42	0.5	0.7414	0.6963	0.8018	0.6874	0.3126	0.7453	0.4989	0.8264
P2-T	All week-4 features	Native categorical handling with explicit Missing category		Calibrated CatBoostClassifier (sigmoid)	Split 60/20/20 with threshold selection on validation; 5-fold CV summary on training	loss_function=Logloss, eval_metric=AUC, depth=6, learning_rate=0.05, iterations=500, auto_class_weights=Balanced, random_state=42	42	0.3	0.7294	0.6521	0.9145	0.5639	0.4361	0.7614	0.5044	0.8264

Table 5: Held-out test-split performance for the four reported configurations after threshold selection on the validation split. P1 is the taught logistic-regression baseline, P2 is the self-learned CatBoost pipeline, and the tuned rows apply the validation-selected threshold of 0.3. Raw CatBoost is the best probability model, while calibrated CatBoost at threshold 0.3 is the best intervention-oriented configuration.

Appendix A: Cross-validation summary on the training split

Appendix A reports the supplementary 5-fold cross-validation summary for the uncalibrated LR and CatBoost models.

Appendix A: Cross-validation summary on the training split

model	roc_auc_mean	roc_auc_std	log_loss_mean	log_loss_std
CatBoost	0.8262	0.0049	0.4989	0.0056
Logistic Regression	0.8155	0.0045	0.5141	0.0047

Appendix A: Five-fold cross-validation was run on the training portion only for uncalibrated logistic regression and CatBoost. Final headline metrics still come from the locked validation/test workflow.

Appendix B: Reusable artifacts note

- Reusable artifact: artifacts/week4_feature_table.parquet
- Grain: one row per (code_module, code_presentation, id_student)
- Cutoff: only day 0 through day 27 information is included
- Leakage exclusions: no date_unregistration, no post-day-27 evidence, no assessment score in the main model

Appendix C: References

[1] L. Kemper, G. Vorhoff, and B. U. Wigger, “Predicting student dropout: A machine learning approach,” *Eur. J. High. Educ.*, vol. 10, no. 1, pp. 28–47, 2020, doi: 10.1080/21568235.2020.1718520.

[2] M. R. Marcolino *et al.*, “Student dropout prediction through machine learning optimization: Insights from Moodle log data,” *Sci. Rep.*, vol. 15, Art. no. 9840, 2025, doi: 10.1038/s41598-025-93918-1.

[3] J. T. Hancock and T. M. Khoshgoftaar, “CatBoost for big data: An interdisciplinary review,” *J. Big Data*, vol. 7, Art. no. 94, 2020, doi: 10.1186/s40537-020-00369-8.

[4] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, vol. 70, 2017, pp. 1321–1330.

[5] A. J. Bowers and X. Zhou, "Receiver operating characteristic (ROC) area under the curve (AUC): A diagnostic measure for evaluating the accuracy of predictors of education outcomes," *J. Educ. Stud. Placed Risk (JESPAR)*, vol. 24, no. 1, pp. 20–46, 2019, doi: 10.1080/10824669.2018.1523734.

[6] O. B. Deho, C. Zhan, J. Li, J. Liu, L. Liu, and T. D. Le, "How do the existing fairness metrics and unfairness mitigation algorithms contribute to ethical learning analytics?," *Br. J. Educ. Technol.*, vol. 53, no. 4, pp. 822–843, 2022, doi: 10.1111/bjet.13217.